

ENGINEERING PORTFOLIO

Ashiqul Islam (Nayeem)

Aerospace & Defence Engineering

niaain10@gmail.com — +1 (204) 887-6259 — <https://www.linkedin.com/in/ashiqul-islam-nayeem-eit-157a79146/>

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Technical Summary

Systems-oriented aerospace engineer experienced in modeling, analysis, and verification of complex mechanical and aerospace systems. Skilled in translating engineering theory into validated, system-level results.

Core Competencies

- **Systems Engineering & Requirements Management:** Lifecycle-driven design, MBSE, requirements traceability, and risk management
- **System Architecture & Trade Studies:** Concept development, performance optimization, sensitivity analysis, and architectural evaluation
- **Control Systems & Dynamic Modeling:** Time- and frequency-domain analysis, controller design, and stability assessment
- **Structural Dynamics & Vibration Analysis:** FEA, modal analysis, structural optimization, and lightweight design considerations
- **Propulsion & Aerothermal Systems:** CFD, thermodynamic modeling, performance testing, and flow conditioning studies
- **Modeling, Simulation & Computational Analysis:** MATLAB/Simulink, Python, XFOIL, and analytical modeling for performance prediction and optimization
- **Verification, Validation & Test (V&V, T&E):** Experimental testing, data reduction, and correlation with analytical and numerical models
- **Operational & Training Support:** Simulator scenarios, operational concept development, SOPs, and procedure documentation
- **Technical Communication & Documentation:** Engineering reports, SOPs, technical manuals, and stakeholder briefings
- **Cross-Functional Collaboration:** Multidisciplinary team coordination, customer and government liaison (e.g., DND), and program support
- **Data-Driven Decision Making:** Metrics analysis, performance assessment, and continuous improvement initiatives (Lean, Six Sigma, Kaizen)

Space Telescope Shipping Container — Systems Design and Dynamic Validation- Mech 4800(Capstone Project with Magellan Aerospace)

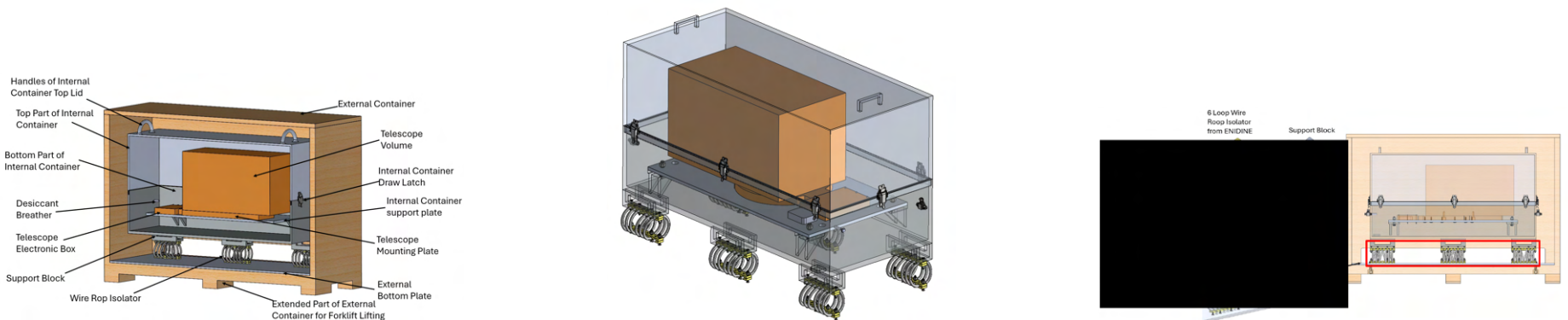


Figure 1: Space telescope shipping container system design and dynamic performance validation

What: Designed a reusable ground-transport container for a high-value, vibration-sensitive space telescope for Magellan Aerospace, ensuring structural integrity, contamination control, and compliance with Mechanical Ground Support Equipment (MGSE) requirements during a 48-hour over-the-road shipment.

How: The design was developed and validated using a requirements-driven engineering process, combining analytical modeling, numerical verification, and risk-based assessment:

- Translated client MGSE requirements into measurable engineering metrics and verification criteria
- Developed a sealed 6061-T6 aluminum internal container with welded construction, gasketed lid interface, grounding provisions, and nitrogen purge capability
- Designed a multi-axis vibration isolation system using wire-rope isolators sized through analytical shock and frequency-domain vibration calculations
- Performed finite element analysis (stress, deformation, buckling, and factor of safety) on critical load-bearing components to validate structural margins
- Conducted a Failure Modes and Effects Analysis (FMEA) to identify transport, handling, and environmental risks and define mitigation controls
- Produced detailed CAD models, assembly procedures, and handling concepts aligned with aerospace ground support best practices

Results: Delivered a fabrication-ready design package including detailed CAD, FEA verification, isolator sizing, assembly procedures, and a complete bill of materials. The final design met 12-inch drop shock, random vibration PSD, and sine dwell requirements while maintaining low transmitted accelerations, high structural safety margins, and a controlled nitrogen-purged internal environment within project cost constraints.

Engineering Focus: Structural and dynamic analysis, vibration and shock isolation, aerospace ground support equipment design, requirements traceability, verification and validation (V&V), and risk-informed engineering decision-making.

Quadruped Robot Dog — Integrated Mechanical, Electrical, and Control Systems Design- Mech 3212(Electrical)

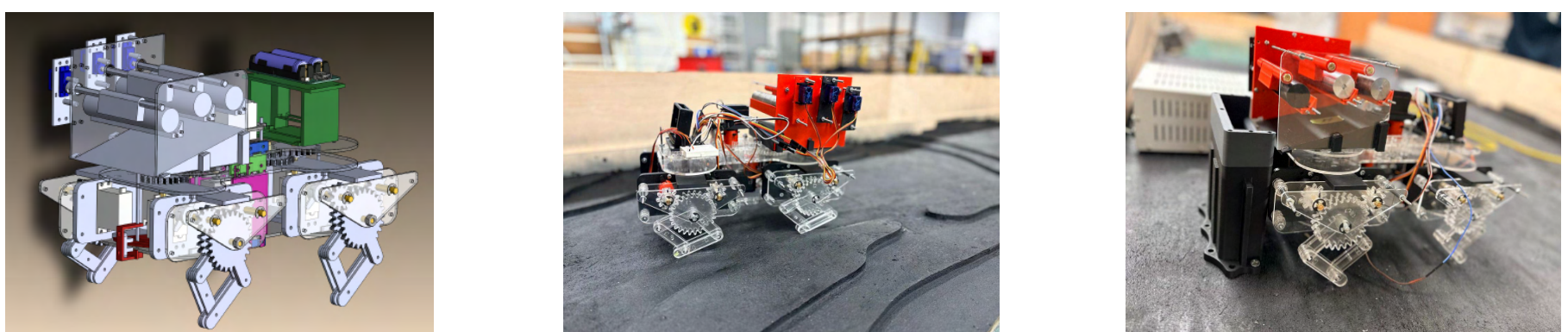


Figure 2: Robot dog with payload delivery

What: Designed and built a four-legged electromechanical robot capable of walking, steering, sensing its environment, and delivering payloads within strict hardware, power, size, and manufacturing constraints. The project required full integration of mechanical design, electrical systems, and embedded control software to create a functional robot dog prototype.

How: The robot was developed through a multi-phase design process combining CAD-based mechanical design, electrical architecture development, and embedded software implementation:

- Designed a pantograph-style four-bar leg mechanism with hip and knee actuation to generate stable sagittal-plane walking motion.

- Performed motion and torque analysis in SolidWorks to validate gait trajectory and determine required gear reductions.
- Designed custom servo-driven gearboxes and a rigid chassis to transmit loads and minimize torsional deflection.
- Integrated an embedded control system using a Micro:bit and Kitronik 16-servo controller to coordinate leg motion, steering, and payload release.
- Implemented walking gaits, turning logic, and payload sequencing in MakeCode Python.
- Designed and integrated a regulated 2S battery system with defined safety and emergency removal procedures.
- Added sensor interfaces (photo-interrupters, planned range and line sensors) to enable closed-loop steering and future autonomous navigation.

Results: The final prototype demonstrated stable forward walking, step climbing, and reliable payload delivery. Mechanical performance validated the leg geometry and gearbox design, while the control software successfully synchronized eight leg servos and multiple auxiliary actuators. Steering performance revealed limitations caused by backlash and manufacturing tolerances, which were identified and documented for future refinement. The project established a complete electromechanical system architecture and a clear path toward fully autonomous navigation.

Engineering Focus: Electromechanical system integration, robotic mechanism design, embedded control systems, servo actuation, sensor integration, prototyping and testing

Aircraft Wing Design & Optimization- Mech 3520(Aerodynamics)

What: Designed and analyzed an aircraft wing for structural strength, weight optimization, and improved aerodynamic performance.

How: CAD modeling of spars, ribs, and skin; FEA for bending, torsion, shear; CFD simulations; iterative optimization for lift-to-drag ratio and weight reduction.

Results: Optimized wing improved lift-to-drag ratio, structural stresses within allowable limits, reduced weight, and enhanced flow uniformity.

Engineering Focus: Structural design, aerodynamic analysis, FEA, CFD, trade-off studies.

Aircraft Materials Selection & Manufacturing- Mech 4192(Aircraft Materials)

What: Selected optimal material and manufacturing process for a hollow circular beam under 4000 N flexural load.

How: Calculated minimum modulus and tensile strength, applied Rule of Mixtures for composites, evaluated fiber-matrix compatibility, minimized weight.

Results: Selected Carbon Fiber-Reinforced Epoxy (weight 0.122 kg), rejected aluminum, steel, and S-glass composites.

Engineering Focus: Material selection, composite mechanics, lightweight design, manufacturing process.

Turboprop Propulsion System Design & Analysis- Mech 4200(Gas Turbine & Propulsion System Project)

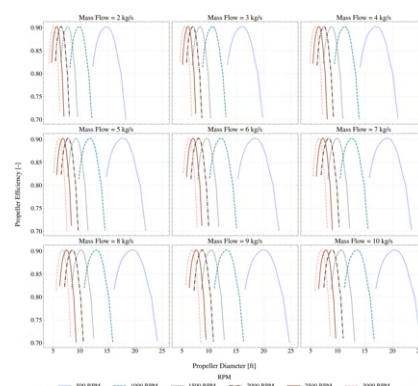
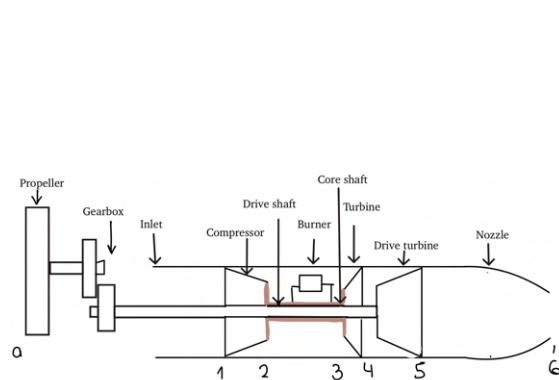


Figure 3: Gas turbine cycle and propeller performance evaluation

What: Designed and analyzed a turboprop system to meet shaft power and thrust requirements efficiently.

How: Brayton-cycle modeling, propeller matching, parametric performance studies using MATLAB/Python.

Results: Achieved target shaft power, high propeller efficiency, acceptable propeller tip Mach numbers.

Engineering Focus: Gas turbine cycle analysis, turboprop sizing, propeller aerodynamics.

Table 1: Operational Turboprop Engine Comparison (Redacted)

Parameter	PT6A-66	Garrett TPE331	PT6A-140	Project Design
Power (HP)	■	■	■	■
Diameter (ft)	■	■	■	■
Blade Count	■	■	■	■
Prop RPM (max/cruise)	■	■	■	■
Prop Eff. (%)	■	■	■	■

Miniature Turbojet Engine Performance Analysis- Mech 4200(Gas Turbine & Propulsion System Lab)

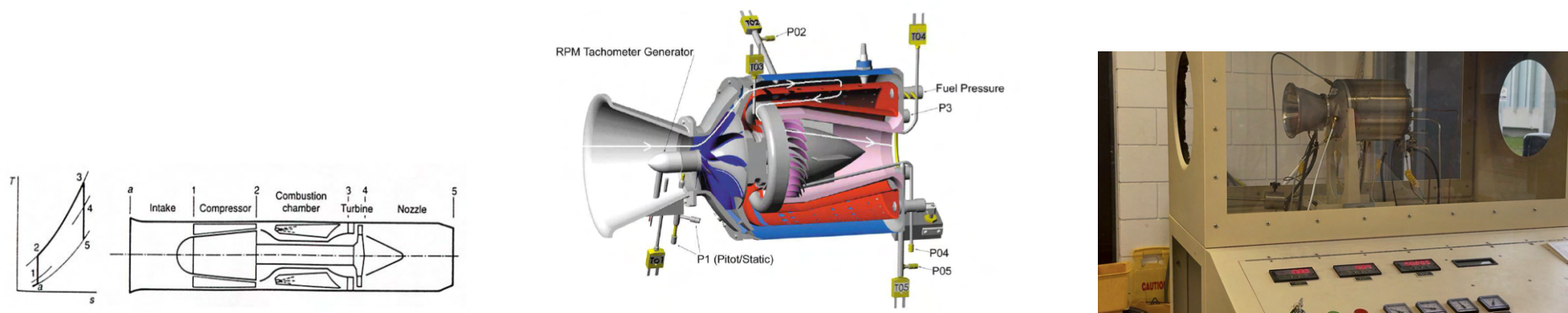


Figure 4: Miniature turbojet engine performance evaluation

What: Evaluated performance of a miniature turbojet engine in laboratory conditions.

How: Measured pressures, temperatures, fuel flow, and RPM; applied thermodynamic calculations for thrust, SFC, and efficiency.

Results: Quantified deviations from ideal cycle predictions; identified component inefficiencies.

Engineering Focus: Propulsion testing, thermodynamics, data analysis.

Aircraft Landing Gear Vibration Analysis-Mech 3420(Vibrations & Acoustics)

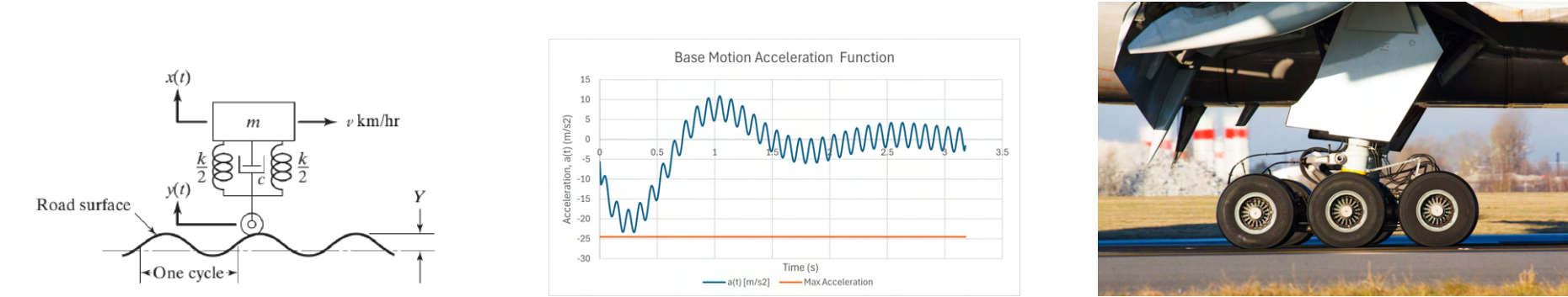


Figure 5: Experimental vibration testing and modal analysis of landing gear system

What: Analyzed dynamic behavior of landing gear system to identify natural frequencies, damping ratios, and mode shapes.

How: Impact hammer testing with accelerometers; MATLAB used to compute frequency response functions and extract modal parameters.

Results: Dominant vibration modes identified; experimental modal properties validated analytical models.

Engineering Focus: Structural dynamics, experimental modal analysis, MATLAB signal processing.

Rear Fuselage Stringer Design — Structural Analysis- Mech 4182(Aircraft Structure)

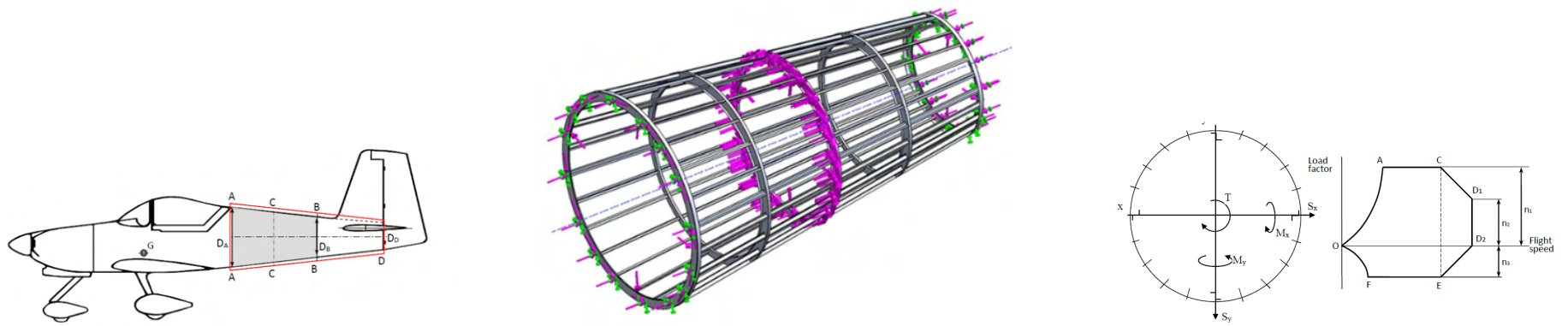


Figure 6: Rear fuselage geometry, loading, and stringer stress distribution

What: Performed structural analysis and preliminary sizing of rear fuselage stringers under combined bending.

How: Modeled fuselage as thin-walled structure with booms; computed section properties and bending stresses.

Results: All stringers met allowable stress; critical section identified.

Engineering Focus: Aircraft structures, thin-walled beams.

Classical Control Systems Design- Mech 3430(Measurement & Control System)

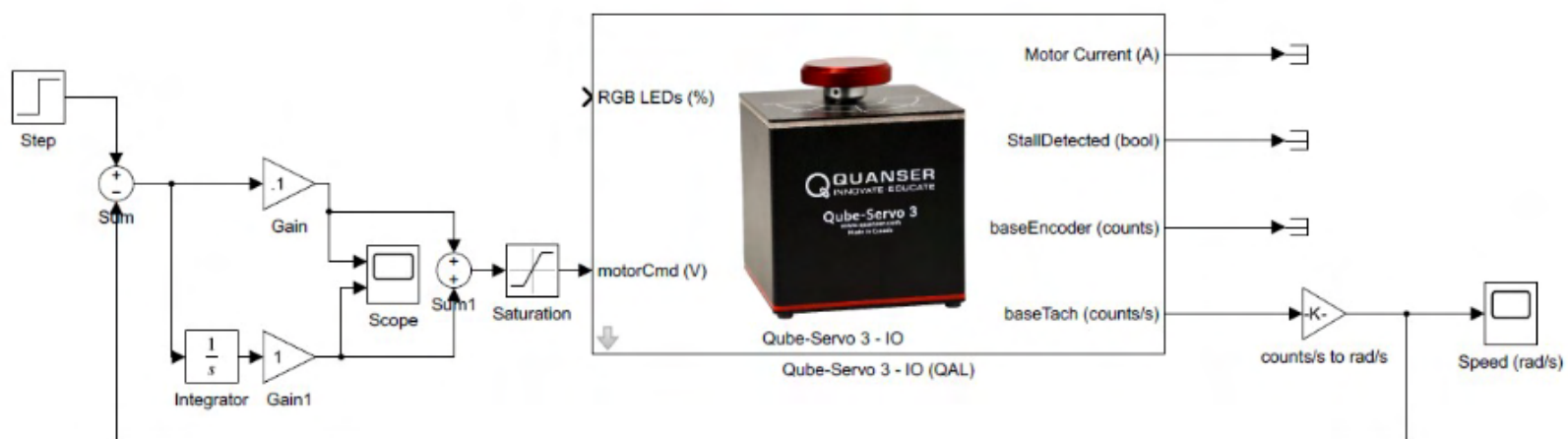


Figure 7: Control system design and frequency-domain validation

What: Designed feedback controllers to satisfy stability, transient, and steady-state requirements.

How: Developed transfer functions, designed PI,PD and PID/lead-lag controllers, evaluated frequency-domain margins.

Results: Stable closed-loop performance achieved with improved accuracy and response.

Engineering Focus: Control systems, root-locus, frequency-domain analysis.

Hovercraft Design & Maximum Performance Evaluation- Mech 3992(Thermofluid Lab)

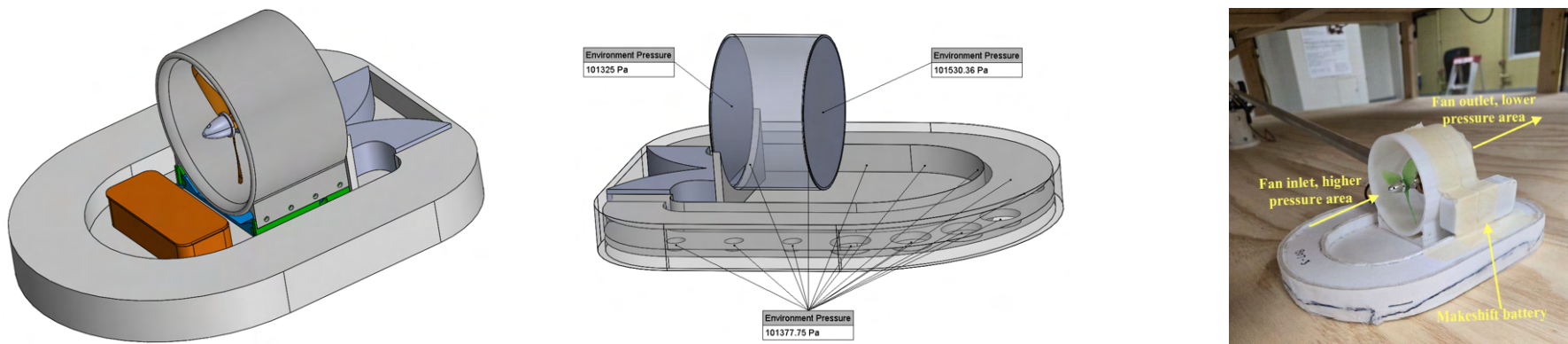


Figure 8: Hovercraft design & analysis

What: Optimized a miniature hovercraft to increase velocity while maintaining lift stability under fixed propulsion constraints.

How: A structured engineering approach was applied, integrating analytical modeling, numerical simulation, and experimental validation to guide design decisions and verify performance:

- Developed analytical models using Bernoulli's equation and Newton's Second Law to determine optimal skirt flow rate, lift height, and minimum lift area.
- Designed and implemented a curved (M-shaped) flow splitter to reduce internal flow losses and redirect airflow toward thrust generation.
- Performed CFD analysis in SolidWorks Flow Simulation to evaluate velocity fields, pressure distribution, and internal flow trajectories.
- Executed controlled static and dynamic experiments to measure thrust, velocity, and hovercraft stability, maintaining consistency with Phase 1 test conditions.
- Conducted trade studies on skirt exit sizing and mass distribution to balance thrust, lift, and stability under operational constraints.

Results: The optimized design achieved a 63% increase in maximum velocity, improving from 4.93 m/s to 8.07 m/s, validating the effectiveness of the redesigned flow splitter and airflow redistribution strategy. CFD results predicted a 28–38% increase in outlet airflow velocity, reaching 18.32 m/s, and confirmed the reduction of internal flow losses and splitter-induced vortices. Experimental testing revealed that an 8.2% increase in system mass led to a reduction in thrust from 0.59 N to 0.30 N, aligning with analytical predictions and highlighting critical trade-offs between speed, thrust, and stability. These findings established clear design limitations and informed recommendations for future optimization.

Engineering Focus: System-level fluid dynamics analysis, analytical–numerical–experimental correlation, performance trade-off evaluation, and design optimization under constraints.

Structural Stress Analysis & Design Optimization — Beams and Trusses- Mech 3502(Stress Analysis)

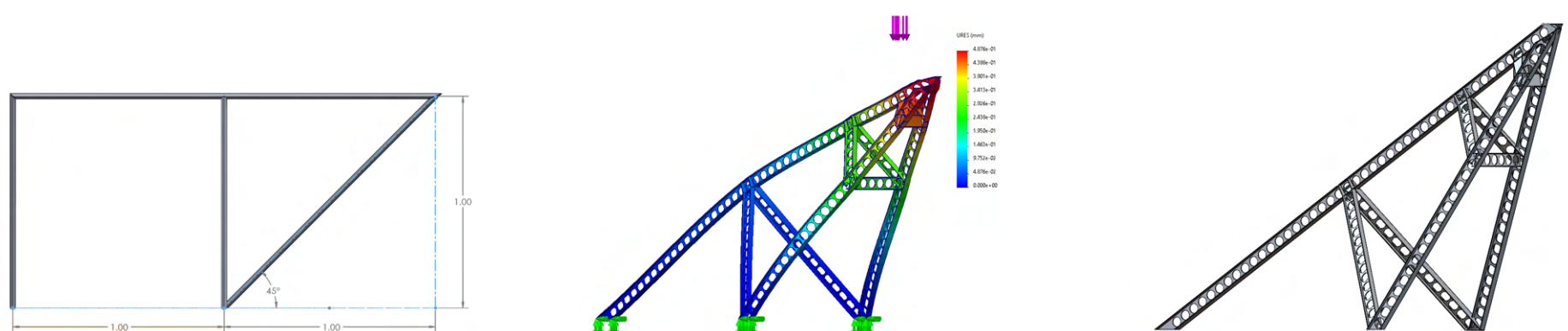


Figure 9: Stress and internal force analysis of beam and truss systems

What: Optimized beams and trusses for material efficiency.

How: Classical mechanics, free-body diagrams, iterative design optimization.

Results: Critical members identified; optimized dimensions improved material efficiency.

Engineering Focus: Structural analysis, mechanics of materials, optimization.

UV Water Purification Tank — CFD-Based Fluids Design- Mech 3492 (Fluid Dynamics)

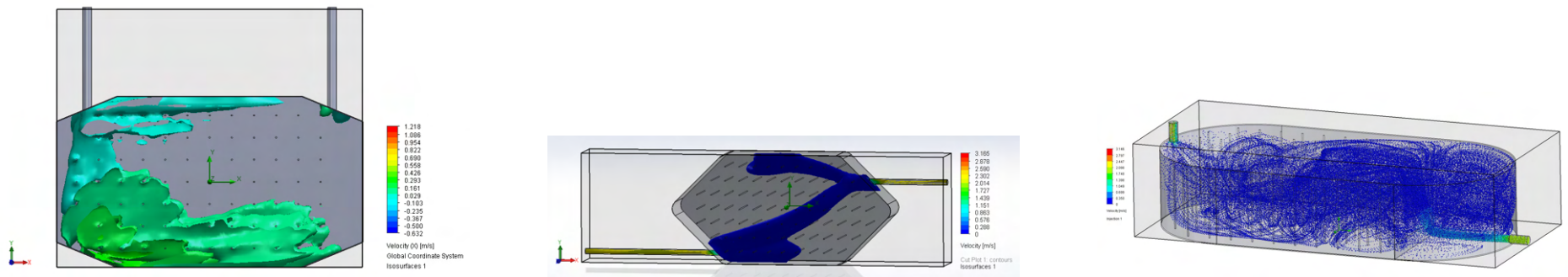


Figure 10: UV purification tank geometry and particle residence-time analysis

What: Designed and optimized a UV water purification tank to minimize volume while ensuring sufficient pathogen exposure time under fixed flow and piping constraints.

How: A requirement-driven design and analysis process was applied, combining CFD-based flow modeling, design iteration, and pump selection to evaluate performance and guide design decisions:

- Defined system requirements including minimum residence time (>400 s), fixed mass flow rate (15 kg/s), inlet/outlet pipe diameter constraints, and material selection.
- Developed and evaluated multiple tank geometries through CFD particle studies to quantify residence time distributions, velocity fields, and pressure gradients.
- Iterated designs to mitigate vortices and short-circuiting by incorporating circular end caps and inlet flow-conditioning features.
- Implemented a wall-block inlet design to reduce inlet velocity and promote uniform flow through the UV exposure region.
- Performed residence time statistical analysis (average, standard deviation, minimum) to rank design concepts objectively.
- Calculated system pressure losses and selected a suitable centrifugal pump using manufacturer pump curves to meet flow and head requirements.

Results: The final wall-block UV tank design achieved an average residence time of 558.6 s, exceeding the minimum exposure requirement, while maintaining a compact tank volume of 8.40 m^3 . CFD particle tracking confirmed reduced inlet short-circuiting, improved flow uniformity, and controlled pressure gradients that promoted extended water retention within the UV exposure region. Comparative analysis across three design iterations demonstrated clear performance trade-offs between residence time consistency, tank volume, and component count, leading to the selection of the optimal configuration. Pump sizing calculations identified a centrifugal pump operating at approximately 74% efficiency at the required flow rate and head, validating system-level feasibility and integration.

Engineering Focus: Fluid system design, CFD-based residence time analysis, requirement-driven design trade studies, flow conditioning, and pump selection for integrated system performance.

Lightweight Mars Rover Structural Optimization — FEA and Experimental Validation- MECH 3982(Mechanical Design)

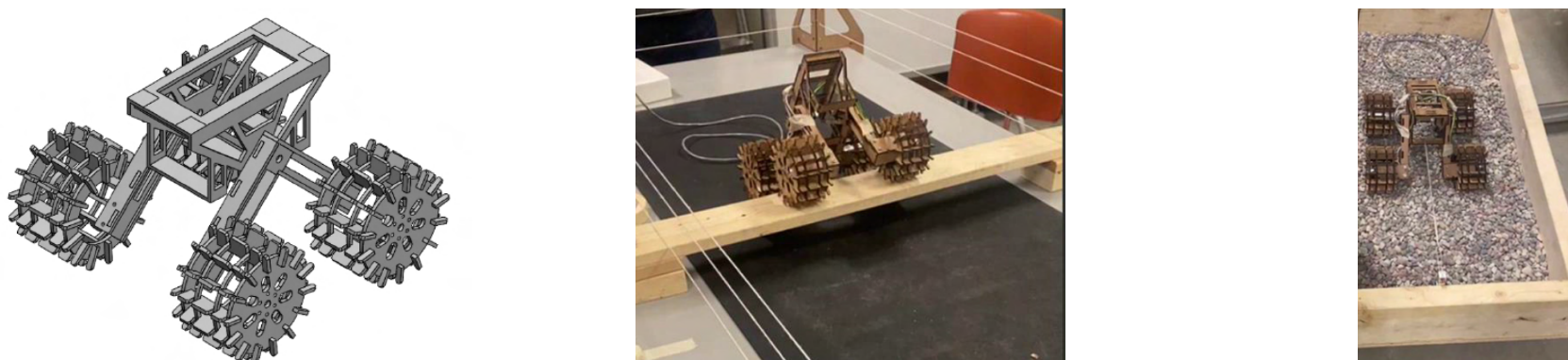


Figure 11: FEA of optimized rover structures, correlated with load and obstacle-climbing tests to verify strength and mobility

What: Redesigned and optimized a ground rover chassis, rocker arms, and wheels to reduce system mass while maintaining structural integrity and improving mobility under defined load and terrain constraints.

How: A systematic, multi-method engineering approach was applied to balance structural performance, weight reduction, and terrain mobility:

- Established quantitative design requirements including maximum allowable deflection (< 0.1 in) under a 7 lb static load, obstacle climb height, and aggregate pulling capability.

- Applied structural optimization principles by introducing truss-inspired cutouts in the chassis and rockers to reduce mass while preserving stiffness.
- Performed detailed finite element analyses (FEA) in SolidWorks on the chassis, rocker arms, and wheels using shell elements, realistic boundary conditions, and gravity loading.
- Conducted mesh convergence studies for all critical components to ensure numerical accuracy and reliable stress and displacement predictions.
- Iteratively refined component geometry based on stress concentrations, displacement locations, and topology study results.
- Redesigned wheel geometry by increasing diameter and paddle height to improve traction and obstacle negotiation while managing rotational inertia.
- Validated analytical and numerical predictions through experimental static load testing, obstacle climbing tests, and aggregate pulling trials on the full rover assembly.

Results: The optimized rover achieved a 37% reduction in total mass, decreasing from 1.875 lb to 1.175 lb, while maintaining structural performance under a 7 lb static load. Experimental testing measured a maximum vertical deflection of 0.033 in, significantly below the allowable limit, confirming adequate stiffness despite aggressive material removal. Obstacle-climbing capability improved substantially, with the rover overcoming a maximum vertical obstacle height of 2.28 in, and up to 2.95 in when using an angled approach ramp, compared to 1.54 in and 1.97 in respectively in the baseline design. Finite element results showed reduced stresses and displacements across the chassis and wheels, with all components remaining well below the material yield strength. Aggregate pull force remained comparable to the original design (2.08 lbf vs. 2.10 lbf), revealing a performance trade-off driven by front-end weight distribution rather than traction alone. These results highlighted the effectiveness of structural optimization while identifying system-level limitations and future improvement opportunities.

Engineering Focus: Structural optimization, finite element analysis with convergence verification, experimental validation, system-level trade-off evaluation, and mobility-focused mechanical design under real-world constraints.

Technical Tools, Methods, and Standards

Engineering Software & Computational Tools

- **CAD & Design:** SolidWorks (advanced CAD, FEA, motion studies, flow simulation), AutoCAD (technical drawings, system layouts)
- **Simulation & Analysis:** ANSYS (structural stress, vibration, modal analysis), MATLAB & Simulink (dynamic modeling, control system design, signal processing), XFOIL (airfoil aerodynamics)
- **Programming & Automation:** Python (engineering computation, data analysis, automation scripts), Maple (analytical modeling)
- **Project Coordination & Documentation:** Oracle, e-Builder, CAPP, Microsoft Office Suite, L^AT_EX

Systems Engineering, Analysis, and Verification

- Requirements-driven Systems Engineering and Model-Based Systems Engineering (MBSE) for concept development, design traceability, and verification planning
- Trade studies, sensitivity analysis, and performance optimization for structural, aerodynamic, propulsion, and fluid systems
- Failure Modes and Effects Analysis (FMEA) for mechanical assemblies, transport systems, and operational risk mitigation
- Verification, Validation, and Test (V&V, T&E) integrating analytical models, numerical simulations, and experimental correlation
- Finite Element Analysis (stress, deformation, buckling, modal response) with rigorous mesh convergence and validation
- Computational Fluid Dynamics (CFD) for flow conditioning, pressure loss analysis, and particle residence-time studies
- Experimental testing and data reduction, including modal vibration testing, structural load testing, and propulsion performance characterization
- Control system modeling, stability analysis, and controller design using time- and frequency-domain methodologies
- Structural optimization for lightweight, high-performance design with manufacturability considerations

Engineering Standards and Regulatory Frameworks

- **NASA Systems Engineering Handbook:** Lifecycle processes, requirements flow-down, verification planning, and risk management
- **MIL-STD-499 / MIL-STD-881:** Systems engineering management, work breakdown structures, and program planning
- **DO-178:** Software assurance awareness within aerospace system development environments